Lily Oglesby

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Education

University of Southern California Los Angeles, CA
Master of Business Research, Marshall School of Business *Aug 2025 – Dec 2026*

– **Upcoming Fall 2026 coursework:** DSO 510 Business Analytics, DSO 529 Regression Analysis, DSO 547 Spreadsheet Modeling, DSO 567 Game Analytics, and DSO 581 Supply Chain Management; Tableau and SAS coursework in progress.

University of California, Berkeley Berkeley, CA
Bachelor of Arts in Astrophysics and Data Science *Aug 2021 – May 2025*

Research Experience

Space Sciences Laboratory, UC Berkeley Berkeley, CA
Undergraduate Researcher *Aug 2021 – May 2025*

– Built reusable Python pipelines for three years of satellite data, applying time-series preprocessing, statistical filtering, and visualization to identify event-related patterns and support a peer-reviewed publication.

Institute for Astronomy, University of Hawai'i at Manoa Honolulu, HI
Research Intern *Jun 2024 – Aug 2024*

– Applied anomaly detection to approximately 120,000 images, trained a convolutional neural network for multiclass classification, and used GPU parallelization to accelerate preprocessing and model execution.

Selected Data Science and Engineering Projects

Interactive Portfolio Game
JavaScript, HTML/CSS, Canvas, GSAP, Howler.js, Tiled *2026*

– Built and debugged a responsive portfolio website presented as a browser game, with an explorable map, collision detection, turn-based battles, mobile controls, animation, audio, asset preloading, and Netlify deployment.

Astronomy Data Analytics Portfolio
Python, pandas, NumPy, scikit-learn, PyMC, Data APIs, Astropy *2024*

– Built reproducible workflows using satellite observations, stellar time-series data, telescope images, and astronomical spectra; applied database queries, feature engineering, regression, cross-validation, Bayesian estimation, high-dimensional preprocessing, and model evaluation.

PM2.5 Health Impact and Forecasting
Python, pandas, scikit-learn, Causal Inference *2024*

– Analyzed CDC and NOAA data to study PM2.5 exposure and asthma outcomes and predict pollution from weather; compared generalized linear and random forest models using leakage-resistant temporal validation.

PostgreSQL vs. MongoDB Yelp Benchmark
PostgreSQL, MongoDB, Python, Jupyter, Docker *2024*

– Designed equivalent workloads across relational and document databases and compared data models, SQL queries, aggregation pipelines, indexes, query plans, scan volume, and runtime performance.

Teaching and Mentoring Experience

USC DSO Research Experiences for Undergraduates Program Los Angeles, CA
Graduate Student Instructor and Mentor *Jun 2026 – Present*

– Led sessions on responsible AI in research and advised students on graduate applications, research statements, and technical presentations.

Introduction to General Astronomy, UC Berkeley Berkeley, CA
Reader and Grader *Aug 2022 – Dec 2024*

– Evaluated assignments across multiple semesters and provided feedback on quantitative reasoning, scientific interpretation, and written solutions.

Mauna Kea Scholars Program, University of Hawai'i Honolulu, HI
Research Mentor *Jun 2024 – Aug 2024*

– Mentored a high school student in Linux, data analysis, research design, anomaly-detection tools, and technical presentation skills.

Python Programming in Astronomy, UC Berkeley Berkeley, CA
Reader, Grader, and Teaching Assistant *Jun 2023 – Aug 2023*

– Led office hours and evaluated Python assignments for correctness, code quality, debugging approach, and clarity.

Skills

Programming and Analysis: Python, SQL, JavaScript, HTML, CSS, pandas, NumPy, scikit-learn, Matplotlib, Jupyter
Databases and Development: PostgreSQL, MongoDB, Git, GitHub, Linux, Docker, VS Code, Canvas API, GSAP, Howler.js, Tiled, Netlify, LaTeX
Methods: Machine Learning, Regression, Random Forests, CNNs, Anomaly Detection, Time-Series Analysis, Bayesian Modeling, Causal Inference, Cross-Validation, Data Visualization